Self-Dual BKZ on NTRU: Dimension-Onset, Fatigue, and a Cross-Engine Reference-Free DSD Observable

Brendan Chambers-Bourgeois

Independent Researcher, Melbourne, Australia

Abstract

The Root Hermite Factor (RHF) is the standard one-number summary of lattice-reduction quality, and the companion paper showed on LWE-Kannan lattices that it is structurally blind to a real difference between BKZ and self-dual BKZ (SD-BKZ). This report measures what that blind spot hides on *NTRU*, where it matters most: near the fatigue point, where reduction stops grinding toward a shortest vector and instead discovers the dense secret sublattice. We report three empirical results, each traceable to per-seed data. First, the SD-BKZ advantage on circulant NTRU is not a smooth function of dimension: tied with BKZ for $n \leq 67$, it spikes at $n \approx 71$–73 (mean up to $+9.7$ nats at $\approx 27\times$ the plateau standard deviation), then settles into a tight $+1.5$–1.7-nat plateau for $n \geq 79$. Second, the defensible NTRU observable is *reference-free*—signed d(LN) has no canonical fixed point on NTRU and is a reference artifact—and under it SD-BKZ reaches dense-sublattice discovery (DSD) at lower modulus than BKZ, the gap growing from $\approx 0\%$ at $n \leq 79$ to 28% at $n = 113$, at $\beta = 20$. Third, a second engine—the G6K sieve—reproduces the RHF-blindness thesis on a fundamentally different oracle and sharpens its scope: at a matched $\beta = 40$ the engines agree on the DSD onset at $n = 89$ (no SD-vs-BKZ gap—an honest null at this dimension), but the gap reopens at $n = 101$ to 9% on the fplll oracle (SD 257.3 / BKZ 280.7, transition row $q = 271$: SD 36/40 vs BKZ 13/40), with the cross-engine question at $n = 101$ still open. What separates the engines at $n = 89, \beta = 40$ is instead the tail: under SD-BKZ the sieve leaves the shortest Gram–Schmidt vector $\approx 4.9\times$ longer than enumeration on identical instances (median log-norm 1.844 vs 0.255, $+1.59$ nats). We also re-validate, on a new modulus family, an fplll GSO numerical fix. All experiments are reproducible via a containerised, determinism-gated benchmark.

Keywords: lattice reduction, BKZ, SD-BKZ, NTRU, overstretched, fatigue, dense sublattice discovery, Rankin profile, sieving, G6K

1 Introduction

NTRU lattices fail differently from random ones. Past the fatigue point [DvW21], reduction stops grinding toward a shortest vector and instead discovers the dense secret sublattice wholesale—and where that transition sits, and which reduction algorithm crosses it first, is exactly the kind of question security estimation answers with models scored by the Root Hermite Factor (RHF). The companion paper [CB26] showed, on LWE-Kannan lattices, that the RHF is structurally blind to differences between BKZ and self-dual BKZ [MW16] that the Li–Nguyen Rankin profile distance d(LN) [LN20] makes visible.

Email: brendanchambersbou@gmail.com (Brendan Chambers-Bourgeois)

Date of this document: 2026-06-13.

This report asks two follow-up questions a data paper is well suited to answer: *(i) does the SD-BKZ phenomenology generalise beyond LWE to NTRU?* and *(ii) does it reproduce on a fundamentally different reduction oracle—sieving (G6K) rather than enumeration?* Both answers turn out to be yes—each with a qualification the data forces, and one that only a reference-free observable can see.

Our contributions are primarily empirical and infrastructural:

- A characterisation of the SD-BKZ advantage on circulant NTRU lattices [DvW21], including a sharp dimension-onset transition (Section 4).
- Evidence that the appropriate NTRU observable is reference-free, and a *dense-sublattice-discovery (DSD) onset* whose SD-vs-BKZ gap grows with dimension as q approaches the fatigue point (Section 5).
- A cross-engine study (Section 6): G6K wired as a second engine behind a reproducibility-gated seam, a self-dual pump-and-jump construction validated against fplll's SD-BKZ, and confirmation that RHF-blindness is engine-independent but regime-dependent—consistent with independent observations of RHF-equality under a non-exact oracle [Row24].
- A criterion-sensitivity audit of the DSD flag itself: the loose b_1-only test over-fires on uniformly-reduced sub-fatigue bases, manufacturing an apparent cross-engine onset gap that the proper two-part criterion eliminates (Section 6). We document the trap because it cost us a headline claim, and it will cost others.
- An independent re-validation of an fplll GSO catastrophic-cancellation fix on a new (NTRU) modulus family (Section 7).

This is a *technical report / data paper*: the emphasis is on the experimental apparatus, the data, and reproducibility, rather than on new theory. Closest prior work is Deng and Jia [DJ24], who give a sieving-based self-dual BKZ that puts a G6K Pump oracle in place of enumeration and thereby attains comparable or better output quality at substantially lower running time, and who observe the SD-vs-BKZ quality gap narrowing as the block size grows. Our aim is complementary rather than competing: we do not propose a faster oracle but a finer empirical characterisation, measuring the SD-vs-BKZ structural difference through the reference-free profile observable across a fixed-β, q-swept regime. White [Whi26] studies dynamic block-size strategies within the same Li–Nguyen framework, complementary to the fixed-β, q-swept regime characterised here.

2 Background

2.1 Circulant NTRU lattices and fatigue

We use the circulant NTRU construction of Ducas and van Woerden [DvW21]. For a prime n and modulus q, sample ternary secrets $f, g \in \mathbb{Z}_q[x]/(x^n - 1)$ (uniform $\{-1, 0, 1\}$, coordinate variance $\sigma^2 = 2/3$) with f invertible, set $h = g/f \bmod q$, and form the $2n \times 2n$ basis

$$B = \begin{pmatrix} qI_n & 0 \\ H & I_n \end{pmatrix},$$

where H is the circulant matrix of h. The secret (f, g) is an unusually short vector in the row span; recovering it breaks the key. Two structures distinguish B from a random q-ary lattice: a *q-vector head* (rows of qI_n that survive reduction as length-q vectors) and a *dense secret sublattice* spanned by the rotations of (f, g). A BKZ-reduced NTRU profile therefore follows the *ZGSA* (Z-shaped) model—a flat head at $\log q$, then a GSA slope—rather than the single-slope GSA line of a random q-ary lattice.

As q grows at fixed n, the secret sublattice becomes relatively denser and easier to detect: at the *fatigue point* $q_{\text{fat}}(n) \approx 0.004\, n^{2.484}$ [DvW21] the attack transitions from the standard regime (secret found only through full SVP) to the *overstretched* regime, where reduction discovers the dense sublattice (*dense-sublattice discovery*, DSD) well before solving SVP. The geometry of this transition—and how self-dual reduction moves through it—is the object of this report.

2.2 BKZ, SD-BKZ, and the two engines

Block Korkine–Zolotarev (BKZ) reduction repeatedly calls an SVP oracle on length-β projected sublattice windows, sweeping left to right. Self-dual BKZ (SD-BKZ) [MW16] alternates a forward (primal) pass with a backward pass on the reversed dual basis, so each tour reduces the basis from both ends. On LWE-Kannan lattices the companion paper [CB26] found the two produce bases of indistinguishable Root Hermite Factor yet measurably different Rankin profiles; this report asks how that difference behaves on NTRU and across two oracle implementations.

We use two engines. **fplll** supplies an exact enumeration SVP oracle; its self-dual variant (`SD_VARIANT`) is our ground truth. **G6K** [ADH+19] supplies an approximate *sieving* oracle through the pump-and-jump BKZ tour (`pump_n_jump_bkz_tour`); we realise SD-BKZ on it as a primal pump-and-jump pass followed by a dual pass run under G6K's `dual_mode` (operations on the dual basis, bounds reflected about full_n/2). The two oracles differ fundamentally—deterministic enumeration versus randomised sieving—which is what makes a phenomenon that survives both engines credible. Sieving carries a fixed setup cost that makes small β uninformative, so all cross-engine experiments use $\beta = 40$, the smallest block size at which the sieve does non-trivial work; this is also the regime in which related q-ary forgery attacks operate [DEP23].

2.3 The reference-free DSD observable

The companion paper's metric is $d(\text{LN}) = \frac{1}{m}\sum_i |r_i - \rho_i|$, where r_i is the slope-removed Gram–Schmidt log-norm profile and ρ_i is the Li–Nguyen BKZ fixed point [LN20]—a GSA line valid for a *random q-ary* lattice. NTRU is not random q-ary (Section 2.1): its reduced profile follows the ZGSA Z-shape, for which no closed-form Li–Nguyen fixed point exists. Comparing an NTRU profile to the LWE line therefore conflates structure with reduction dynamics, and the *signed* advantage $d(\text{LN})_{\text{BKZ}} - d(\text{LN})_{\text{SD}}$ can flip sign under a different reference—it is a *reference artifact.*

Two observables survive this. First, the *unsigned* profile divergence $\frac{1}{m}\sum_i |r_i^{\text{BKZ}} - r_i^{\text{SD}}|$ is reference-free (it never names ρ); on NTRU it is large precisely at the transition and reproduces under three independent references. Second, and what we report throughout, is a binary *dense-sublattice-discovery* flag—a *two-part* criterion: a reduction has found the secret sublattice when its profile has collapsed onto it, $\#\{i : \log \|\mathbf{b}_i^*\| < \log \sqrt{2n \cdot 2/3} + 0.5\} \le n+1$, *and* the shortest GS vector clears the floor, $b_1 = \min_i \log \|\mathbf{b}_i^*\| > 1.5$. The count threshold is the q-independent ternary-secret log-norm $\log \sqrt{2n \cdot 2/3}$ ($= 2.39$ at $n = 89$; the empirically measured b_1 saturation value is $\approx$1.96) plus a 0.5-nat margin, evaluated per n (2.75 at $n = 67$ to 3.01 at $n = 113$; Section 5 reports the sensitivity to freezing it). The $b_1 > 1.5$ half alone is necessary but not sufficient—it over-fires on uniformly-reduced sub-fatigue bases (Section 6). Both parts are properties of the output basis alone, independent of any modelled reference. The onset q is where the per-seed DSD rate crosses 50%, linearly interpolated between adjacent q-grid cells (`extract_dsd_onset.py`); the SD-vs-BKZ onset *gap* is the object of Sections 5 and 6.

3 Experimental Setup

3.1 Generators and engines

NTRU bases are built by `generators/ntru.py` following Section 2.1; each basis is checked per seed before any reduction (dim $= 2n$ and the key relation $Hf \equiv g \pmod{q}$; the qI_n block layout is covered by unit tests). Both engines sit behind a single seam: `run_single` takes a `backend` argument, and `make_backend` returns either an fplll or a G6K backend exposing a common `tour()`/`gso()` interface. The per-tour reduction loop is then engine-blind—the same loop drives enumeration and sieving, so the two cannot diverge through duplicated control flow. The G6K self-dual tour is a primal `pump_n_jump_bkz_tour` followed by a dual pass executed inside a `temp_params` block that flips `dual_mode`; this was validated against fplll's `SD_VARIANT` on a random q-ary lattice ($n = 80$, $\beta = 40$, three tours) by checking that the SD−BKZ profile-change signature (the GS-slope flattening) agrees in direction and magnitude across engines (fplll -0.00039 vs G6K -0.00040 per tour); its 50-tour NTRU behaviour is separately corroborated by the cross-engine soundness check of Section 6.

3.2 Reproducibility and determinism gates

Both engines run in containers built from a digest-pinned base with `snapshot.debian.org` apt pinning; all source-built engine binaries (G6K and the patched fplll) fix a `-march=x86-64-v3` target, and the canonical fplll path ships the version-pinned `fpylll` 0.6.4 wheel—identical bytes on every machine. Each result is a per-seed JSON carrying the full input parameters and output profile; a SHA-256 manifest gates byte-identity. The fplll path is held to numerical reproducibility (10^{-4} tolerance on reference seeds) by `verify.sh` and to byte-identity by the SHA-256 seed manifest; the G6K path to an exact SHA-256 match by `verify_g6k.sh` against a captured reference hash, confirmed on a second machine (the CI runner).

Determinism on the sieve required care (ADRs 005–008). G6K's multi-threaded sieve has a nondeterministic reduction race, so all sieving runs single-threaded (`threads` $= 1$); across-seed parallelism still uses all cores, since seeds are independent. The fplll global RNG must be seeded both before basis construction and again before the sieve (G6K samples from it), and the `SieverParams` seed is a no-op that we never rely on. With these, multi-tour G6K runs are bit-identical run-to-run, and—verified at $n = 80$, $\beta = 60$, three tours—per-tour re-seeding is itself a no-op: the construction seed alone determines all N tours, so the policy is to seed once at construction. The new modulus and engine introduced no schema change to the seed JSON, preserving the companion paper's SHA baselines.

3.3 Campaigns and parameter grids

Every experiment is a named campaign in `config/sweep.toml`, fixing generator, engine, n-grid, β, q, tour count, and seed count so a single command reproduces a cell. The dimension-onset sweep (Section 4) fixes $q = 97$, $\beta = 20$, 50 tours and sweeps $n \in [59, 89]$ at 20 seeds each. The fatigue and DSD-onset sweeps (Section 5) fix n and sweep q across the fatigue point. The cross-engine sweeps (Section 6) fix $n = 89$, $\beta = 40$, 50 tours and sweep a matched q-grid on *both* engines—the G6K campaign (`ntru_g6k_qsweep`) and the fplll campaign (`ntru_xeng_qsweep`) write to leaves that share the same $(q, \text{precision}, \text{tours}, n, \beta)$ key, so a comparator reads the two engines cell-for-cell. The Kahan re-validation (Section 7) runs $n = 127$ at $q \in \{971, 1087, 1201\}$ through a separate patched-fplll campaign whose seeds land in their own tree, never overwriting the canonical seeds.

Table 1: SD-BKZ advantage on NTRU: tied → onset spike → plateau.

n	mean adv	std	seeds> 0
59–67	~ 0	tight	3–9/20
71	+4.35	5.59	16/20
73	+9.68	5.21	19/20
79–89	+1.5–1.7	0.20–0.33	20/20

Table 2: Reference-free DSD-onset (proper two-part criterion): SD-BKZ reaches dense-sublattice discovery at lower q than BKZ, the gap growing with n. Onsets are 50%-rate crossings, linearly interpolated between grid cells; gap% is computed from these unrounded onsets (denominator: SD onset). Bracketing cells carry $N = 100$ seeds at $n \geq 89$ and $N = 20$ at $n = 67/79$ (where the gap is ≈0); the $n = 113$ endpoint comes from the completed precision ladder.

n	SD onset q	BKZ onset q	gap%
67	144.6	145.4	1
79	171.2	171.0	0
89	238.0	283.3	19
101	428.6	512.2	20
113	729.2	930.4	28

4 The SD-BKZ Advantage Has a Dimension-Onset on NTRU

On circulant NTRU at $q = 97$ ($\beta = 20$, 50 tours; standard regime throughout the sweep, though only marginally so at the smallest dimension: $q_{\text{fat}}(59) \approx 100$), the SD-BKZ-vs-BKZ advantage has a *sharp* dimension transition, not a smooth ramp (20 seeds per n):

Three regimes: tied for $n \leq 67$; an onset zone at $n = 71$–73 where the mean spikes to +4–10 nats with ≈27× the plateau standard deviation; and a stable +1.5–1.7-nat plateau for $n \geq 79$ (std 0.20–0.33, 20/20 wins). The onset variance is not noise but bimodality: the per-seed advantage at $n = 71$ ranges from −6.7 to +13.4 ($n = 73$: −5.1 to +16.0)—SD-BKZ cracks the dense sublattice on most instances and loses outright on the rest. The onset spike exceeds the plateau—non-monotonic (Figure 1; Table 1). One scale caveat: the NTRU $d(\text{LN})$ baseline (≈50) sits roughly an order of magnitude above the LWE-Kannan scale (final $d(\text{LN})$ ≈4–5 at the companion paper's central cells), so advantage magnitudes are comparable within NTRU only—sign and structure, not raw nats, carry across to the companion paper.

5 Fatigue and the Cross-Dimension DSD-Onset Gap

Before sweeping q we confirm the metric is sound here. The $n \approx 71$–73 divergence of Section 4 is reference-robust: re-derived from the stored profiles under three independent references (Li–Nguyen, GSA-zero, and the reference-free $\frac{1}{m}\sum_i |r_i^{\text{BKZ}} - r_i^{\text{SD}}|$), it remains large under all three (32.7 nats reference-free at $n = 73$ vs 0.03 at $n = 67$). The *signed* advantage, by contrast, is a reference artifact on NTRU (Section 2.3); accordingly everything below reports the reference-free DSD observable, never the signed quantity.

Sweeping q across the fatigue point at fixed n, the reference-free BKZ-vs-SD-BKZ profile divergence peaks near $q \approx q_{\text{fat}}$. The SD-BKZ DSD-onset occurs at progressively *lower* q than the BKZ onset as n grows—the gap widens with dimension:

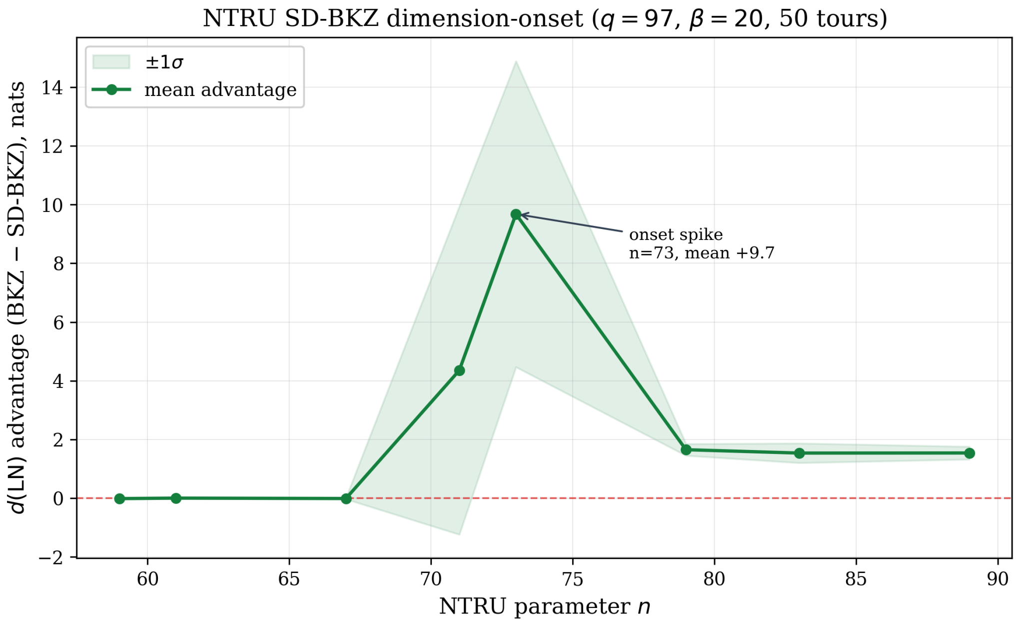

Figure 1: NTRU SD-BKZ dimension-onset ($q = 97$, $\beta = 20$, 50 tours, 20 seeds per n, $\pm 1\sigma$ band; in the bimodal onset zone $n = 71$–73 the band is a spread indicator only, not a distributional summary). The signed advantage is reference-robust at the spike (Section 2.3); shown here for the dimension transition. Recomputed from the per-seed `advantage` field.

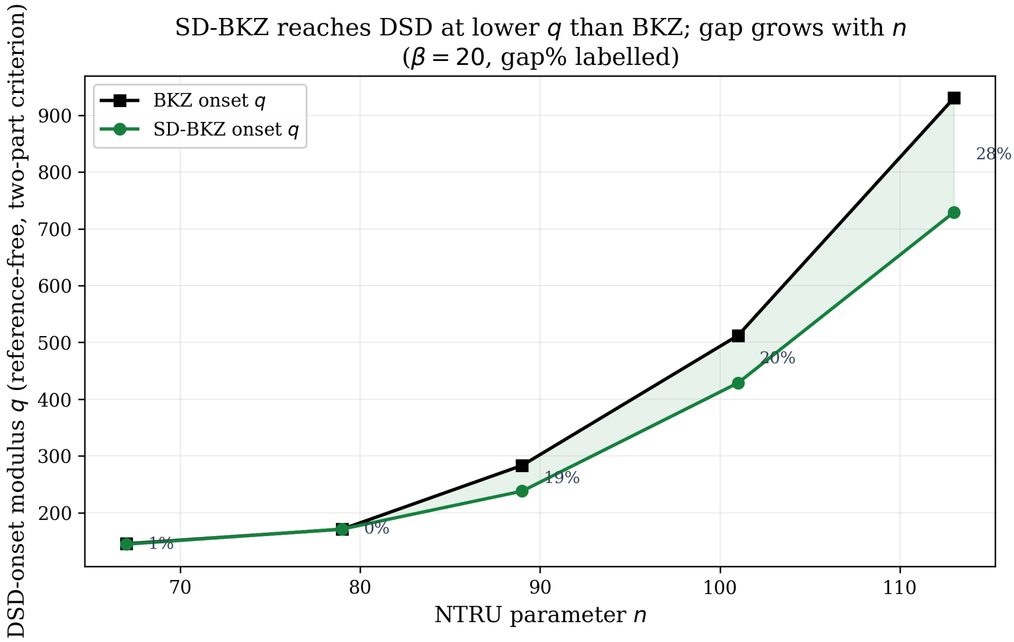

Figure 2: Reference-free DSD-onset modulus (proper two-part criterion, Section 2.3) for SD-BKZ vs BKZ as a function of n ($\beta = 20$), with the SD-vs-BKZ gap% labelled. SD-BKZ reaches dense-sublattice discovery at progressively lower q than BKZ; the gap grows from ≈0 to 28%. Values mirror Table 2.

The extraction evaluates the short-count threshold per n (Section 2.3); freezing it at its $n = 89$ value 2.888 across the grid shifts only the near-zero-gap rows ($n = 67$: 166.6/167.5; $n = 79$: 181.7/182.5; $n = 113$ BKZ 925.0, gap 27%) and leaves the gap trend unchanged—the trend is threshold-robust. $n = 127$ is excluded (fplll GSO catastrophic cancellation [CB26, §8], re-validated in Section 7, plus an off-grid crack). The widening is shown in Figure 2.

6 Cross-Engine: SD-BKZ on a Sieve Oracle

The G6K engine seam and the self-dual pump-and-jump construction are described in Section 3.1; recall that the construction was validated against fplll's `SD_VARIANT` by matching the per-tour GS-slope-flattening signature (−0.00039 vs −0.00040). This section reports what the validated sieve engine then shows.

RHF blindness is engine-independent but regime-dependent. On NTRU at $q = 97$, $\beta = 40$, the companion paper's thesis reproduces on the sieve: at $n = 67/101/113$ the stored Hermite-factor values of the G6K BKZ and SD-BKZ bases are equal seed-for-seed (to machine precision at $n = 67$, bit-identical at $n = 101/113$; the fplll diffs are $\leq 2 \times 10^{-5}$) while $d(\mathrm{LN})$ differs. The effect is regime-dependent: at $n = 89$—the cell containing the short-vector events—the values diverge broadly on every seed, fired or not. This is consistent with RHF-equality reported under a non-exact (approximate-enumeration) oracle [Row24]; the present work measures the structure RHF misses.

Cross-engine validation of the construction. A dimension-resolved fplll-vs-G6K comparison at $q = 97$ ($n = 89/101/113$, 20 seeds each, $\beta = 40$, 50 tours) confirms the

Table 3: Cross-engine soundness at $q = 97$ (fired = first GS log-norm < 3.5, agreement screen only): fplll never fires where G6K does not; the jointly-fired $n = 89$ seed matches exactly (2.332), and the single G6K-only fire is the sieve finding a vector enumeration misses. Jaccard 1.00 rows are vacuous (0/0).

n	G6K fired	fplll fired	Jaccard
89	2/20	1/20	0.50
101	0/20	0/20	1.00
113	0/20	0/20	1.00

self-dual G6K construction is sound (Table 3). A run "fires" when its first GS log-norm drops below 3.5—a deliberately loose short-vector screen used only for this agreement check, never as a DSD flag. At $n = 101/113$ neither engine fires (the Jaccard 1.00 is vacuous agreement on the empty set). At $n = 89$ fplll's single firing (seed 19) is corroborated by G6K with the identical first GS log-norm 2.332, and G6K additionally fires on one seed (17) that the deterministic enumeration misses—consistent with a randomised oracle finding a genuinely short vector. fplll's exact oracle never fires where the sieve does not, so the $n = 89$ short-vector events are real but sparse, not an artifact of the approximate oracle.

At a matched $\beta = 40$ the two engines agree: no DSD-onset gap. We run the *same* grid ($n = 89$, $\beta = 40$, 50 tours, 15 moduli $q \in [97, 211]$ including a dense onset window $q \in \{191, 193, 197, 199\}$, 100 seeds per cell) on both the fplll enumeration engine and the G6K sieve, under the *proper* two-part DSD criterion: a basis has reached dense-sublattice discovery when its profile has collapsed onto the dense sublattice, $\#\{i : \log\|\mathbf{b}_i^*\| < \log\sqrt{2n \cdot 2/3} + 0.5\} \le n + 1$, with the threshold $\log\sqrt{2n \cdot 2/3} + 0.5 = 2.888$ at this $n = 89$, *and* $\min_i \log\|\mathbf{b}_i^*\| > 1.5$. The two-part form matters. The looser $b_1 > 1.5$ test alone over-fires on uniformly-reduced sub-fatigue bases, where the tail is cleared but the profile has not collapsed onto the sublattice: read as a DSD flag, it converts the $q = 137$ clearing counts of Table 4 (38 vs 12 enum, 61 vs 23 sieve) into a spurious "matched cross-engine DSD gap"—a claim an earlier draft of this report made, and which the two-part criterion refutes (the audit record is `results/validation/dsd_criterion_sensitivity.json`). Under the proper criterion the cross-engine picture is a clean *null*: at $q = 137$ neither SD-BKZ nor BKZ fires DSD on *either* engine, and the onset is pinned by the dense window at 100 seeds per cell—fplll SD 193.7 / BKZ 193.8, G6K SD 193.1 / BKZ 194.0, a 0% SD-vs-BKZ gap on both oracles (Table 4). The DSD rate rises smoothly through the window (37% at $q = 191$ to 85% at $q = 199$, 100% at $q = 211$, near-identical curves for both variants and both engines), so the null is measurement-resolved, not grid-limited: densifying from $N = 40$ to $N = 100$ moved every onset by less than one modulus unit.

The cross-engine result at $n = 89$ is therefore not a confirmation of the $\beta = 20$ DSD-onset gap of Section 5 but a *regime constraint at that dimension*: the $\beta = 20$ gap does not survive the move to $\beta = 40$ at $n = 89$, and the two-engine picture is a flat null. At $n = 101, \beta = 40$ the picture is different. An 8-prime coarse grid $q \in \{127, 149, 181, 211, 241, 271, 307, 331\}$ at $N = 40$ seeds/cell on the fplll engine (run 2026-06-12, 50 tours, p=250, same two-part criterion as $n = 89$) recovers a 9% SD<BKZ DSD-onset gap (SD 257.3, BKZ 280.7, 50%-rate crossings), with the transition row $q = 271$ showing SD-BKZ firing on 36/40 seeds against BKZ's 13/40 on identical bases—the largest single-q SD-advantage cell in the corpus. Whether the $n = 101$ gap survives cross-engine is a single experiment that has not yet been run (extractor: `extract_dsd_onset.py --n 101 --beta 40`; seeds: `results/seeds/ntru/q{127,...,331}/p250_mt50/n101_beta40`). We therefore treat the matched-β null as established at $n = 89$ and the gap as established on fplll at $n = 101$,

Table 4: Cross-engine at $\beta = 40$ under the proper two-part DSD criterion: both engines reach DSD-onset together ($\approx q = 194$, dense onset window, 100 seeds/cell) with no SD-vs-BKZ gap—an honest null. The separate min-GS-clearing event ($\min_i \log \|\mathbf{b}_i^*\| > 1.5$ alone: aggressive tail reduction, *not* DSD) shows SD>BKZ on both engines and a sieve-vs-enum structural difference ($q = 137$, 100 seeds).

	fplll (enum)		G6K (sieve)	
	SD	BKZ	SD	BKZ
two-part DSD @ q=137	0/100	0/100	0/100	0/100
two-part DSD onset (50%-rate)	193.7	193.8	193.1	194.0
min-GS-clearing @ q=137	38/100	12/100	61/100	23/100
median $\min_i \log \|\mathbf{b}_i^*\|$	0.255	0.104	**1.844**	0.140

with the cross-engine claim at $n = 101$ explicitly open. What the matched-β comparison *does* surface is a structural difference between the oracles themselves. The min-GS-clearing event—an aggressively reduced tail ($\min_i \log \|\mathbf{b}_i^*\| > 1.5$, short of full DSD collapse)—fires more often under SD-BKZ than BKZ on both engines (38 vs 12 on fplll, 61 vs 23 on G6K at $q = 137$, 100 seeds), and under SD-BKZ the G6K sieve leaves the shortest GS vector far longer than fplll enumeration on identical instances: median log-norm 1.844 (sieve) vs 0.255 (enum), +1.59 nats $\approx 4.9\times$ in vector norm (under plain BKZ the difference is small: 0.140 vs 0.104). The sieve leaves a more aggressively-reduced sub-fatigue tail than enumeration. This is a genuine cross-engine observation—both engines start from the byte-identical primal basis (100/100 seeds) and the determinant is conserved (median drift $\sim 10^{-6}$, max 1.3×10^{-4}) (`results/validation/sieve_vs_enum_min_gs_clearing.json`)—but it is aggressive reduction, *not* dense-sublattice discovery, and we label it as such.

The reference-free observable remains essential on these cells: where the min-GS-clearing event fires, the *signed* d(LN) advantage is large and negative on both engines, because an aggressively reduced q-ary tail is far from the LWE-derived d(LN) reference. Report the unsigned profile structure, never the signed advantage.

7 Re-Validating the fplll GSO Fix on a New Modulus Family

The companion paper documents a catastrophic-cancellation bug in fplll's squared-form GSO update at cryptographic moduli, and a Kahan-compensated fix. We re-validate that fix on a new (NTRU) modulus family: on $q \in \{971, 1087, 1201\}$ at $n = 127$, the five contaminated runs (four BKZ, one SD-BKZ) recover from the precision-floor clamp ($b_1 = -345.388$) to -0.160, -0.123, -0.060, -0.070, and -0.113—inside the never-contaminated control band (-0.03 to -0.43), within which the patched control runs also remain—and the patched build passes all 15 fplll regression tests. The patch removes the *catastrophic* cancellation specifically: 208 transient mid-trajectory clamps persist at this dimension and precision (11 shallower than -0.1; 157 in -0.1 to -3.5; 40 deeper, down to -173); raw values are side-logged while the clamped value enters the stored profile. Separately, five of the twelve patched SD-BKZ runs end with b_1 at the -345.388 floor: these are not residual contamination but genuine dense-sublattice cracks—SD-BKZ cracking the circulant off-grid at $n = 127$, the same crack that excludes $n = 127$ from the trend of Section 5—whose collapsed profiles bottom out at the clamp floor. The fix is thus validated on two modulus families (the companion paper's $q = 3329$ LWE, 38%$\to$0% degeneracy, and this NTRU set).

8 Discussion

Three observations bear on how SD-BKZ is modelled in NTRU security estimation. First, the SD-BKZ advantage is *not* a smooth function of dimension: the sharp onset spike at $n \approx 71$–73 (Section 4), exceeding the plateau, means a security estimate interpolated from a coarse n-grid can miss where self-dual reduction is most effective. Second, the quantity that captures the SD-vs-BKZ difference on NTRU is reference-free: the signed $d(\mathrm{LN})$ advantage flips sign under the reference and must not be reported (Section 2.3); the defensible observables are the unsigned profile divergence and the DSD flag. Third, and most consequential, the DSD-onset *gap*—SD-BKZ reaching dense-sublattice discovery at lower q than BKZ—grows with dimension at $\beta = 20$ (Section 5). At matched $\beta = 40$ the enumeration and sieving oracles agree at $n = 89$ (within 1 modulus unit at 100 seeds/cell, Section 6); at $n = 101$ the gap reopens on the fplll oracle to 9%, motivating but not yet establishing a cross-engine claim at that dimension. Where the two oracles differ at $\beta = 40$ is the aggressiveness of sub-fatigue tail reduction—the G6K sieve clears the shortest GS vector markedly more than fplll enumeration on identical instances. A model that treats SD-BKZ and BKZ as RHF-equivalent (as the Root Hermite Factor, blind here, would suggest) misses that at $\beta = 20$ self-dual reduction crosses the overstretched threshold earlier.

These findings sit alongside two lines of work. The $\beta = 40$ regime and the Z-shaped q-ary profile geometry match the small-modulus forgery setting of Ducas–Espitau–Postlethwaite [DEP23], where the same profile structure is the attack surface. And the move away from an RHF-centric view toward sublattice density echoes recursive lattice reduction [AEPSD23], which abandons the Hermite-factor framing for a Rankin/dense-sublattice one—the same direction our reference-free observable points. We do not benchmark against speed-oriented reduction tools such as `flatter` [RH23]: the object of study is basis *structure* at a fixed reduction effort (profile convergence, DSD onset), not attack wall-clock, so the relevant comparison axis is oracle type (enumeration vs sieving), which Section 6 covers.

We are careful about scope. The dimension-onset spike is shown at $\beta = 20$, $q = 97$; the cross-engine comparison at $\beta = 40$, $n = 89$. The signed $d(\mathrm{LN})$ is reported nowhere as a result; and the cross-engine comparison at $\beta = 40$, $n = 89$ is an honest null on the DSD-onset (a regime constraint, not a confirmation), its positive content being the sieve-vs-enum tail-reduction difference; the matched-$\beta = 40$ gap reopens at $n = 101$ on fplll (9%, Section 6), with the cross-engine measurement at that dimension still pending. The criterion-sensitivity episode of Section 6 generalises: a binary structural flag should test the structure (here, profile collapse onto the sublattice), not a single order statistic that aggressive reduction can satisfy without the structure being present. And $n = 127$ is excluded from the onset trend (Section 5): the fplll GSO cancellation [CB26, §8] contaminated those cells, and the circulant crack there is off-grid—it is a validation target (Section 7), not a trend point.

9 Reproducibility

Every result traces to per-seed JSON files under `results/seeds/`, SHA-256-gated for byte-identity and reproducible across machines from digest-pinned containers (Section 3.2). Each experiment is a named campaign in `config/sweep.toml`, so a cell is regenerated by a single `run_campaign` invocation naming the campaign, q, and seed count. The cross-engine soundness records and DSD-onset tables are committed under `results/validation/` with their own reproduction commands; the determinism and self-dual-semantics decisions are recorded in ADRs 005–008. Each figure has one figure module: Figure 1 renders directly from the per-seed data, and Figure 2 mirrors Table 2, whose values reproduce from the seeds via `extract_dsd_onset.py --trend`. The companion paper's SHA baselines are preserved unchanged: the NTRU generator and the G6K engine were added without altering the seed JSON schema.

10 Conclusion

On NTRU, the difference between BKZ and self-dual BKZ that the Root Hermite Factor cannot see is real, reference-free, and consequential: SD-BKZ reaches dense-sublattice discovery at a lower modulus than BKZ, with a gap that grows with dimension at $\beta = 20$. At a matched $\beta = 40$ the enumeration and sieving oracles agree on the onset at $n = 89$—there they differ instead in how aggressively they reduce the sub-fatigue tail, the sieve markedly more than enumeration. At $n = 101$ the gap reopens on the fplll oracle to 9% (SD 257.3 / BKZ 280.7, Section 6); whether it survives cross-engine is open. The companion paper argued the RHF is blind on LWE; this report shows the blindness persists on NTRU and on a sieve, exactly where it matters for the overstretched transition. The apparatus—a determinism-gated, containerised, two-engine benchmark with per-seed evidence—is released so the measurements can be reproduced and extended.

References

[ADH+19] Martin R. Albrecht, Léo Ducas, Gottfried Herold, Elena Kirshanova, Eamonn W. Postlethwaite, and Marc Stevens. The general sieve kernel and new records in lattice reduction. *EUROCRYPT 2019; IACR ePrint 2019/089*, 2019. G6K; pump-and-jump BKZ; dual_mode.

[AEPSD23] Divesh Aggarwal, Thomas Espitau, Spencer Peters, and Noah Stephens-Davidowitz. Recursive lattice reduction, 2023. arXiv:2311.15064. Abandons rHF for dense-sublattice density; DSP via Rankin — conceptual cite for the rHF-blind framing.

[CB26] Brendan Chambers-Bourgeois. SD-BKZ vs. BKZ: A reproducible lattice-reduction benchmark via the Li–Nguyen Rankin profile. Zenodo, 2026. Companion paper (paper 1): *The Root Hermite Factor is Blind: Structural Differences in Lattice Reduction Revealed by the Li–Nguyen Rankin Profile*, contained in the record. Zenodo is its only release — no ePrint or other pre/post-publication. URL: https://zenodo.org/records/20518060, doi:10.5281/zenodo.20518060.

[DEP23] Léo Ducas, Thomas Espitau, and Eamonn W. Postlethwaite. Finding short integer solutions when the modulus is small. In *CRYPTO 2023; IACR ePrint 2023/1125*, 2023. Z-shape q-ary geometry; Mitaka forgery at $\beta \approx 40$ — matches the $\beta = 40$ regime used here.

[DJ24] Xixuan Deng and Huiwen Jia. A new self-dual BKZ algorithm based on lattice sieving. In *Frontiers in Cyber Security (FCS 2023)*, volume 1992 of *CCIS*, pages 323–336. Springer Singapore, 2024. Conference FCS 2023; proceedings published 2024. Self-dual BKZ with a G6K sieve oracle — closest precedent to the cross-engine self-dual construction here. doi:10.1007/978-981-99-9331-4_22.

[DvW21] Léo Ducas and Wessel van Woerden. NTRU fatigue: How stretched is overstretched? In *ASIACRYPT 2021*, 2021. Defines the circulant NTRU construction and the fatigue point $q_{\text{fat}} \approx 0.004\, n^{2.484}$; ZGSA Z-shape profile.

[LN20] Jianwei Li and Phong Q. Nguyen. A complete analysis of the BKZ lattice reduction algorithm. Cryptology ePrint Archive, Report 2020/1237, 2020. URL: https://eprint.iacr.org/2020/1237.

[MW16] Daniele Micciancio and Michael Walter. Practical, predictable lattice basis reduction. In Marc Fischlin and Jean-Sébastien Coron, editors, *EUROCRYPT 2016, Part I*, volume 9665 of *LNCS*, pages 820–849. Springer, Berlin, Heidelberg, May 2016. doi:10.1007/978-3-662-49890-3_31.

[RH23] Keegan Ryan and Nadia Heninger. Fast practical lattice reduction through iterated compression. In *CRYPTO 2023*, volume 14083 of *LNCS*, pages 3–36, 2023. The flatter tool; IACR ePrint 2023/237. Speed-oriented reduction — cited for scope contrast (structure vs wall-clock).

[Row24] Joe Rowell. *Theoretical, Practical, and Engineering Aspects of Lattice Cryptanalysis and TLS Attestation.* PhD thesis, Royal Holloway, University of London, 2024. Ch.3: approximate-enum oracles in BKZ; SD-BKZ achieves the same RHF as BKZ under a non-exact oracle (RHF-blind, independent).

[Whi26] Trill White. Dynamic analysis of lattice basis reduction algorithms. AMSI Summer Research Scholarships 2025–26, Deakin University, 2026. Dynamic block-size strategy within the Li–Nguyen framework. URL: https://srs.amsi.org.au/wp-content/uploads/sites/92/2026/03/white_trill_srs-report.pdf.